No Mountain in the Sentence

A discipline for trusting an analogy exactly as far as it earns — worked on a physical model of large language models

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> *A note on what this is. I'm an engineer, not an ML researcher. This grew out of a project to understand transformer mechanics by building a physical analogy for them — and the most useful thing that came out of it wasn't the analogy, it was the discipline I had to invent to keep the analogy honest. That discipline is the subject here; the transformer model is the worked example that proves it cuts. I expect people with deeper expertise to revise or reject parts of the example. The method is meant to survive that — it's a tool for not being fooled by your own metaphors, and it works best precisely when the metaphor is good enough that you'd otherwise forget to doubt it.*

Abstract

A good analogy is the most dangerous tool in technical reasoning, because it works often enough to earn your trust and then fails silently in the places it can't see. This essay is about the discipline that makes one safe to use anyway. It states the discipline first — three questions and a single rule — and then earns it the hard way, by applying it to a sustained physical analogy for how large language models work and reporting where the analogy holds, where it bends, and where it told a confident lie that nearly became a published result. The mountain is the case study. The method is the point. Read this if you have ever built a metaphor you liked too much to test.

I. The discipline, stated before the evidence

Most writing about analogies arrives at its caveats last, as a list of disclaimers after the picture has done its persuading. That order is backwards, and the reason is not stylistic. **Order controls what a reader defends and what they stay curious about.** Build attachment first and every limitation reads as an attack to be resisted; state the test first and the same limitations read as the test doing its job. So the test comes first here, before you have met the analogy it will be used on, so that you hold it in your hand the entire way down.

The discipline has one rule and three questions.

The rule: state the claim with no mountain in the sentence. Whatever the analogy is — a mountain, a river, a spring, a clockwork — strip it out and say what you are actually claiming about the real system in the system's own terms. If you cannot, you do not have a claim. You have an image, and an image cannot be true or false, which means it cannot be checked, which means it will quietly smuggle in whatever it pleases. The instant a claim is stated without the metaphor, it becomes either obviously sound, obviously wrong, or precisely testable — and all three of those are better than the comfortable fog the metaphor was hiding in.

Then, before trusting the picture on anything new, three questions:

1. **Can the claim be stated without the analogy at all?** (The rule, as a gate. Fail it and stop here.)
2. **Does the claim secretly assume uniformity** where the real system is lumpy and heterogeneous? Analogies love smooth, even materials because they are easy to picture; real systems are usually a few load-bearing parts and a lot of filler, and the metaphor will quietly average over that difference.
3. **Does the real answer depend on a variable the analogy cannot see?** Every analogy renders some properties of its target vividly and is simply blind to others. When the deciding variable lives in the blind region, the picture will not stay silent — it will confidently answer using whatever variable it *can* see, and that substitution is where it lies.

And the conclusion the questions serve: **an analogy can be trusted to point at the right neighborhood, and must never be trusted to name the mechanism, whenever the deciding variable lies outside what it can draw.** It is a generator of hypotheses, not a provider of them. The discipline is what keeps the first role from impersonating the second.

That is the whole method. Everything below is the bill coming due — the cost of proving it on a real analogy rather than asserting it on a clean one.

What's old here, and what isn't

None of the epistemology above is new, and it would be dishonest to imply otherwise. The discipline of testing an analogy by its *critical disanalogies* — the points where source and target differ in a way that breaks the conclusion — is old and well-developed. Philosophers of science have formalized it at least since Mill (1843) and Keynes (1921), and the modern treatment (see the Stanford Encyclopedia of Philosophy's entry on analogical reasoning) lays out essentially these criteria: judge an analogy by the causal relevance of its similarities, by the differences that negate its conclusions, and by whether its argument is not just valid in form but sound in fact. "State the claim with no metaphor in the sentence" is a compression of that last demand — separate the picture's rhetoric from its actual logical structure. I did not discover this. I re-derived it the hard way, from the inside, while trying to keep one specific analogy honest.

So I want to be exact about what this essay offers, because it is narrower than "a new framework" and more useful than "nothing." Three things. **First, a formulation that survives under load** — "no mountain in the sentence" is a slogan you can actually invoke mid-thought, when you are excited about an idea and least inclined to run a six-criterion checklist; the packaging is the point, the way a good mnemonic is. **Second, a worked example in a live domain** — the philosophy literature illustrates with rectangles and boxes; this runs the discipline on a current, load-bearing technical analogy that real people are using right now to reason about real systems, and shows exactly where it bites. **Third, a documented near-miss** — a first-person case of the discipline catching an error I had already committed to, which is the rarest and most convincing thing here precisely because almost nobody publishes the experiments they *almost* ran. The principle is borrowed. The slogan, the live example, and the caught mistake are what I am actually contributing.

II. The analogy, built to be tested

Here is the picture the rest of this essay will prosecute. It is a good one — good enough that you should feel its pull, because a test only means something against an analogy you were tempted to believe.

A large language model is a mountain, and the data running through it is water. The **weights** — the billions of numbers fixed when training ends — are solid rock, a permanent landscape of ridges and valleys. The **activations** — the numbers computed fresh on every input — are water flowing over that rock. **Training** carves the mountain the way real water carves real stone: not by softening it, but by sustained flow — pour enough water over it for long enough and the channels deepen, the valleys settle into shapes that route any future water correctly. **Inference** — using the model — is a single stream run once, far below the volume that moves stone, so it finds its way across a landscape that no longer changes. The difference between carving and merely running over is not a change of material; it is a change of dosage.

The picture extends with surprising reach. **Attention** is the moment water at a ledge draws on the streams beside it, forming a temporary channel that leaves no mark and vanishes when the next input arrives. The **residual stream** is each drop's own riverbed running the length of the range. A **cache** of past computation is the damp trail earlier water left on the rock, so the next drop only computes its own splash. A lightweight **fine-tuning adapter** is a tarp laid over one valley to divert the flow without touching the bedrock beneath.

The spine of the whole thing is one axis: **permanence.** What training carves is permanent; what attention lays down per input is not. And that single distinction — what persists versus what is recomputed — is *exactly* the real difference between weights and activations. That is not decoration on the side of the machine; it is the machine's central architectural fact, wearing a shape you can hold in your head.

It is a genuinely useful picture. Hold it now against the three questions, and watch where it stops being the territory.

III. The test running: three applications and a near-miss

Application 1 — the foundational images (where the picture imports physics the system doesn't obey)

Run the rule on the analogy's own central verb, and the deepest break appears immediately. Every sentence above has water *falling* — descending the range, pouring over the precipice, seeking the low valley. State it with no mountain in it: *is there a force driving the forward pass toward a minimum?* No. A driving force toward a low point would mean the computation *relaxes toward a lowest-energy state*, and that is optimization — it happens only during training, never during inference. The forward pass has no driving force at all; "descending the range" is nothing but applying layer one, then layer two, then layer three. Nothing falls, nothing is minimized. The analogy reached for the cause it could most vividly *draw* — water obviously falls — and in doing so silently fused inference with training, the exact confusion the permanence axis exists to prevent. “Down” means *later in the stack*, not *lower in energy*, and the picture survives only if you hold that translation fixed at every step. This is question 3 in its purest form: the deciding fact (there is no energy gradient) lives in the analogy’s blind spot, so the analogy answered with the variable it could see (gravity), and lied.

Two more foundational imports fail the same way, more quietly. **Conservation:** fluid intuition smuggles in conservation of volume, but the real system conserves nothing — distinguishability is destroyed where streams merge, and magnitude is *created* as each layer writes new signal into the channel. Never reason “what flows in flows out.” **Locality:** “the drop looks at its neighbors” imports adjacency and falloff, but the real operation reaches every permitted source at once with no notion of near or far, and (in the generative models this describes) only ever draws from *earlier* positions, never later. In each case the metaphor rendered something the mechanism lacks — gravity, conserved volume, spatial locality — and you only catch it by stripping the metaphor out and checking.

Application 2 — capacity (where the analogy is half right, and the half it misses is the important half)

Now a subtler result, because it cuts both ways. Does the geometry of a thousand-dimensional space transfer to a picture you can hold in three? Run question 1. The *relational* facts — distance, angle, similarity, projection — transfer exactly; the proof is that cosine similarity is a pairwise operation computed by the identical formula at any dimension. So “water finding the channel carved into the rock” is *honest*: a trained network really does carve a structured landscape, and reasoning about how an input routes through it is sound even though you cannot picture it. That half cuts *for* the analogy, and it is worth saying plainly, because the discipline is not about distrusting the picture everywhere — it is about knowing exactly where.

What does *not* fully transfer is **capacity** — but here precision matters, because the naive version of this claim is too strong, and the discipline is supposed to catch exactly that. The capacity *problem* transfers perfectly, and it’s worth seeing why: the geometry is continuous all the way up. One axis (1D) forces everything onto a line; a second relieves the crowding; a third relieves more — and that same rule, *independent directions relieve crowding*, runs unbroken into 1024 dimensions with no rung where the kind of thing changes. So a 3D terrain isn’t a metaphor for the space; it’s the same geometry at a visible scale. A finite substrate has only so much room to hold distinct things, and once a region is densely carved, new knowledge competes with what’s there rather than slotting in beside it — that is the same fact as catastrophic forgetting, and you can watch it on a terrain. What does not transfer is the *magnitude*: how much room there is, and how the limit arrives. In low dimensions independent directions are scarce, so a 3D surface crowds early; in high dimensions the headroom looks miraculous — but it is worth being exact about its source, because the source of the room is also the source of the interference. Exact orthogonality buys no miracle: d dimensions hold exactly d perpendicular directions. The explosion to exponentially many appears only once you accept *near*-orthogonality — small overlaps instead of clean right angles — and those overlaps are not free; they *are* the interference budget. So the roominess and the crowding are the same fact measured twice: the space fits far more than its axis count precisely because it lets features bleed slightly into each other, and that same bleed is what eventually degrades them. You cannot pack a thousand exactly-perpendicular pencils into a room; you can pack vastly more if each is allowed to lean a little on the others, and the leaning is the cost. So the blind spot is narrow and specific: the picture conveys *that* the space saturates and *what happens when it does* with full fidelity, and misleads only about *how much* room there is before it does. This is question 3 at its most surgical — not "the picture is unreliable about capacity," but "the picture is unreliable about capacity's *magnitude*, and reliable about everything else." The blind spot sits right next to a region of perfect fidelity, which is exactly why the right lesson is never "distrust the picture" but always "distrust it *here, for this specific reason.*"

Application 3 — the deepest wall (the test applied to the hardest case)

The hardest case is the one the picture most wants to be wrong about. Can you look at a trained mountain and read off what concept lives in a given valley? The honest answer is no, and the reason is the deepest limit in the whole subject. Concepts are not stored in clean, separate locations; a single direction in the space participates in many unrelated concepts at once, and any given concept is smeared across many directions. This is the same capacity fact from Application 2, seen from its consequence: because the space holds exponentially many overlapping directions, the model uses them all at once, and nothing sits in a place you could point to. The leading attempt to un-mix this — sparse autoencoders — produced early excitement and is now under serious doubt, with multiple 2025 evaluations finding it failing to beat simple baselines and at least one major lab publicly stepping back from it. The picture's instinct — that you could excavate a clean concept from a clean location — is exactly the thing the mechanism forbids. The mountain cannot show you where a capability lives, because in a superposed system *lives* is the wrong verb. This is the climax of the test: applied to the hardest case, the analogy does not merely bend, it hits a wall the mechanism itself builds.

The near-miss — the test catching a lie I had already believed

The three applications above are the discipline working in advance, on claims I had not yet committed to. This last one is different, and it is the reason I trust the method rather than merely recommending it, because it caught a result I had already designed an experiment around and was excited to run.

The analogy's central claim is that inference runs over frozen stone. I went after it directly. Take a small model, teach it some invented facts, then freeze every layer past a chosen depth so that downstream stretch is mechanically locked. Fine-tune only the upstream layers on conflicting facts until the original facts degrade. Then feed the frozen downstream the internal state it *originally* received, before the fine-tune, and watch the old behavior return — proving the degradation lived in what *reached* the frozen computation, not in the computation itself. Loss without the rock moving. I had the model picked, the task designed, the controls specified. It felt like a clean, fundable little result.

Then I stated it with no mountain in the sentence. If the downstream layers never changed, feeding them the exact input they originally received reproduces their original output *by construction* — same fixed function, same input, same answer, guaranteed, for any model, any task, any depth. The recovery I had built the entire experiment around could not fail. And a result that cannot fail cannot teach you anything; it is an identity wearing the costume of a finding. I had wrapped an elaborate rig around a tautology, and the only thing that caught it was the rule — the same rule stated at the top of this essay, applied to my own work instead of someone else's.

What survives is smaller and real: the degradation half stands alone (a behavior *can* be made to fail with a downstream computation held fixed), and the informative measurement is not whether restoring the *whole* state recovers the behavior — it must — but how little of it must be restored, which is not guaranteed to recover anything and therefore *would* actually measure where the damage concentrates. That would be a real experiment — designed, not yet run — and I am confident enough in the reasoning to stake the method on it: the tautology above is a proof and needs no running, but this measurement is a genuine question whose answer I do not already have. It exists as a question at all only because the seductive version dissolved under a sentence with no mountain in it.

IV. What the test proves — about the analogy, and about the method

Look back at where the picture failed, and the failures are not scattered. **The analogy has teeth in exactly one circumstance: when the question genuinely turns on permanence — what is fixed versus what is fluid, the weights versus the water. It bends or breaks the instant the deciding variable lives somewhere else** — in the geometry of the loss landscape, in which specific components carry the load, in the physics of the hardware, in the dynamics of the optimizer. Gravity broke because the deciding fact was the absence of an energy gradient. Capacity bent — but only on magnitude, because the saturation *problem* transfers fine and only the counting scale has no spatial form. The interpretability wall stands because the deciding fact is superposition. In every case the metaphor stayed confident and answered with the variable it could draw, and the discipline is the only thing that caught the substitution — including the case where the honest verdict was "mostly transfers, fails in one narrow place" rather than a clean break.

So the analogy earns a precise verdict, not a grade: it is a strong *generator* of questions and a weak *provider* of mechanisms. It points reliably at what to interrogate, and it must hand off to real analysis the moment the interrogation turns on a variable it cannot see. Used on that leash, it is genuinely valuable. Off the leash — the moment it starts naming causes — it hands you something elegant and wrong, and it feels like insight right up until someone states the claim without it.

And that is the transferable result, the one that outlives this particular mountain. The method is not about transformers. It is about the structural fact that **every analogy renders some of its target vividly and is blind to the rest, and the blind region is exactly where it will lie to you with the most confidence** — because it cannot see that it is guessing. The three questions are a way of finding the blind region before you trust the picture inside it. The rule — no metaphor in the sentence — is how you force a claim out of the fog and into a form that can be checked. They cost almost nothing, they work on any analogy you are tempted by, and they are most valuable precisely when the analogy is good, because a good analogy is the one you will forget to doubt.

I built the most convincing version of this picture I could, specifically so that breaking it would mean something, and the most useful thing it ever did was tell me, in advance, where to stop trusting it — including once, when it nearly told me a lie I would have published. A weak analogy hides its edges. A strong one shows you exactly where it stops being the territory. The method is how you make it show you.

Written by E. A. Flores, Apiana AI, Inc. The mountain-and-water framework and its full build-out are developed in a companion paper, "The River and the Canyon"; this essay isolates the reasoning discipline that paper applies. I used Claude (Anthropic) extensively as an editorial and technical sounding board across many rounds of drafting and revision — pressure-testing arguments, checking claims, and challenging the structure. The framework, the predictions, the errors, and every structural and editorial decision are mine.